

Le Chalk-Trick

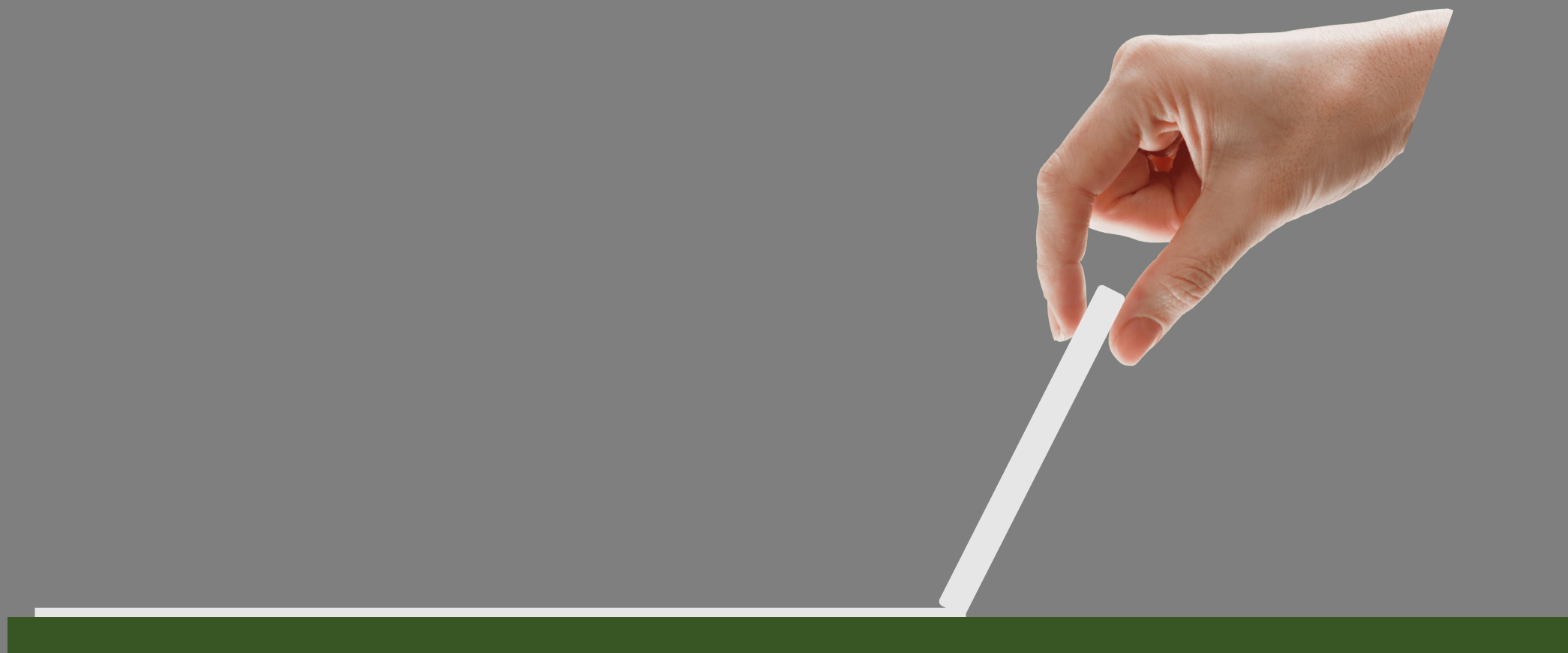


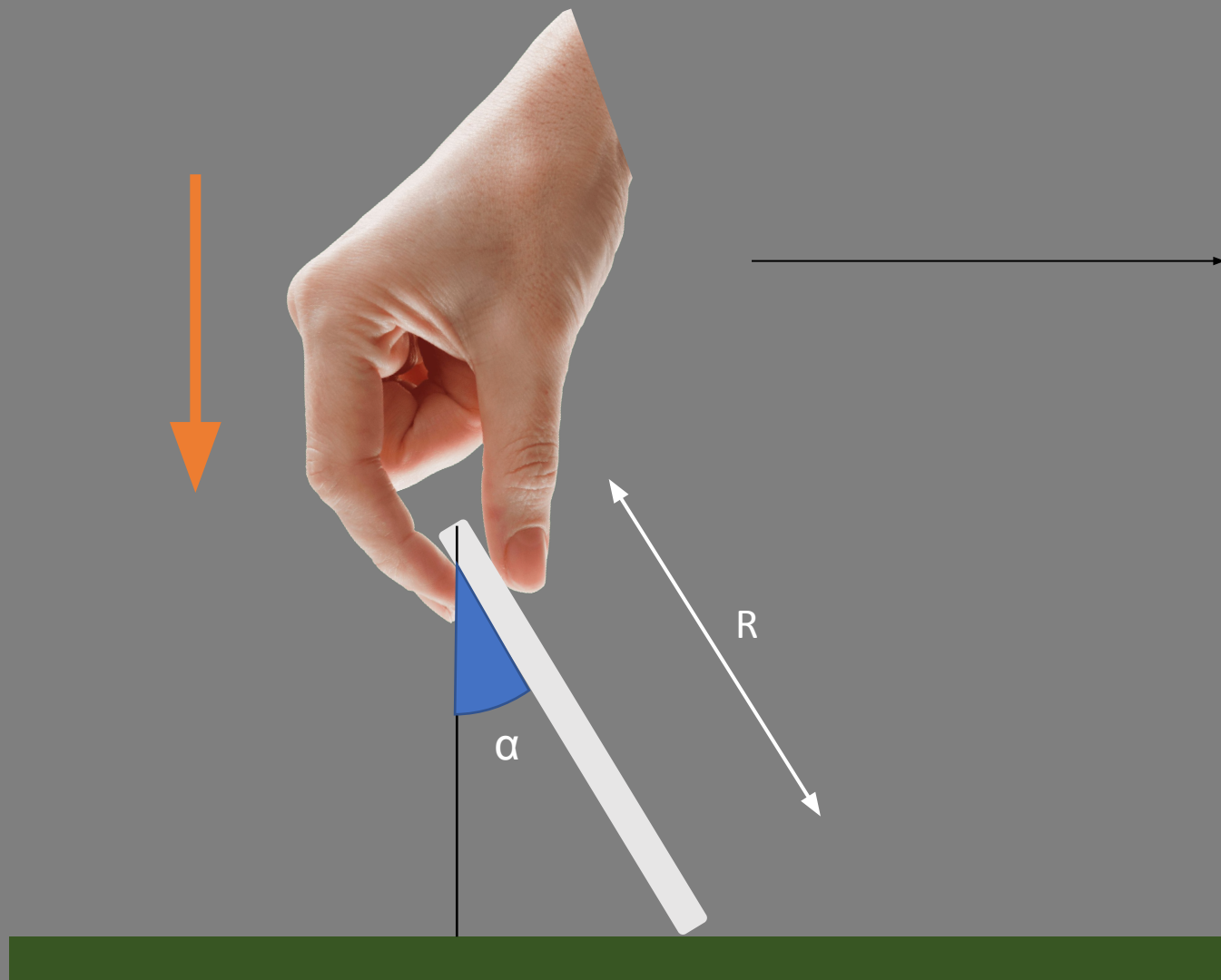
Plan

- I. Intro
- II. Présentation de l'expérience
- III. Résultats de l'expérience
- IV. Partie theorique

I. Intro



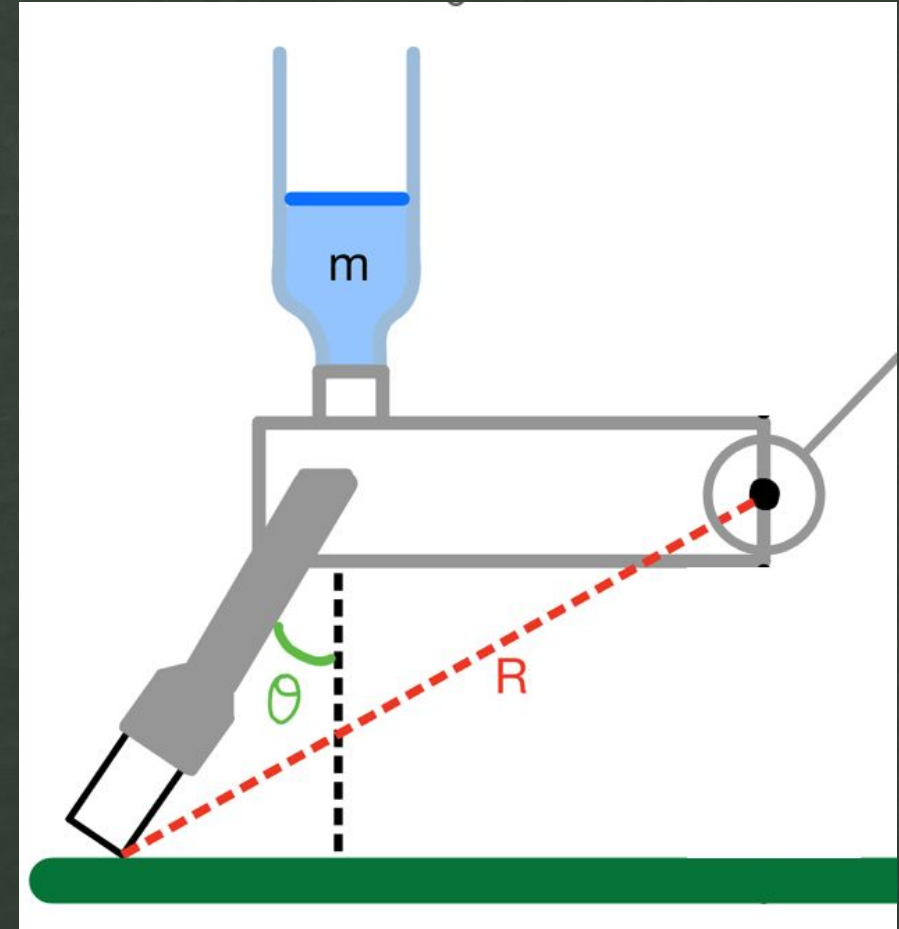
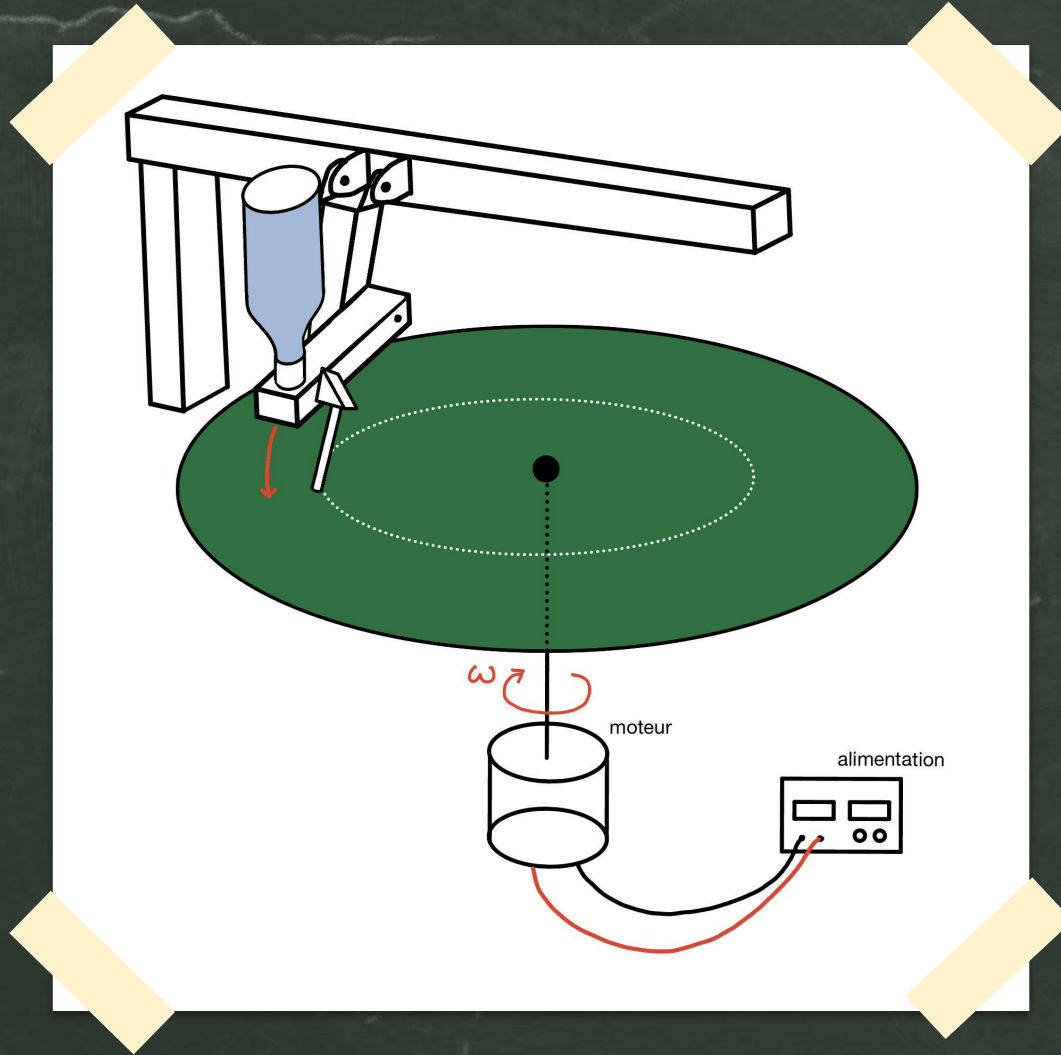




4 paramètres :

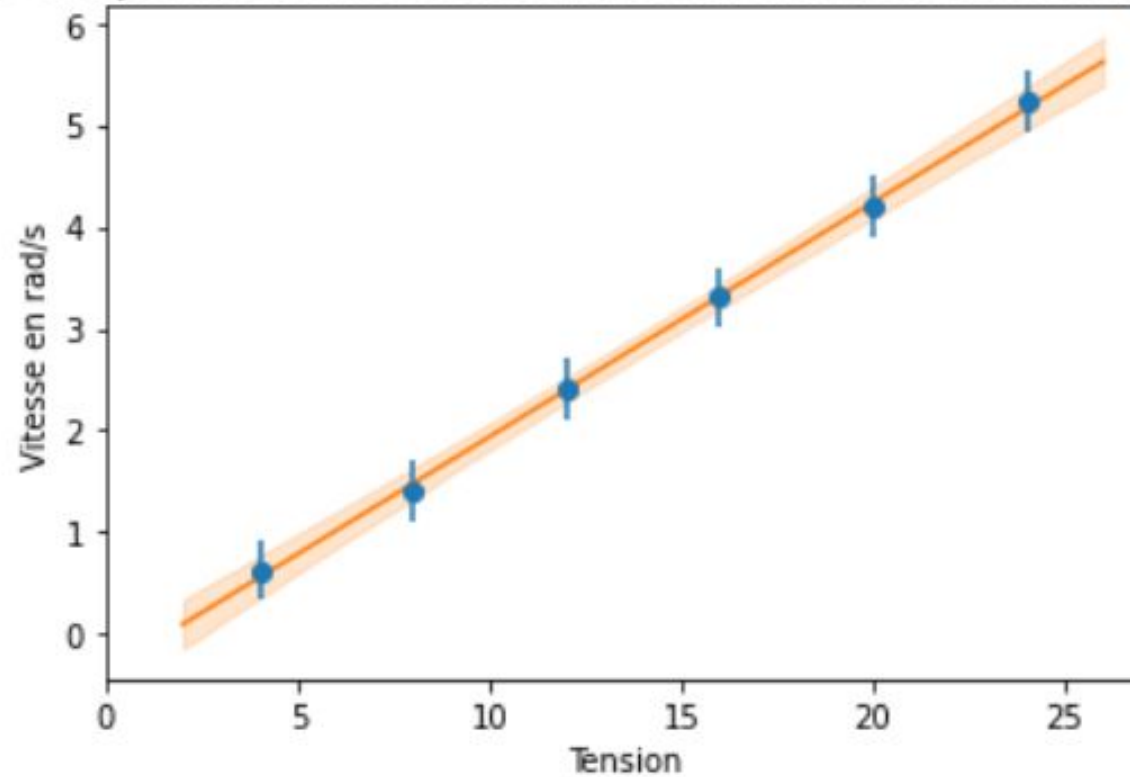
- Longueur R de la craie (dm)
 - Vitesse (m.s-1)
 - Force (0-10 N)
 - Angle α (0-90 degrés)

II. Expérience



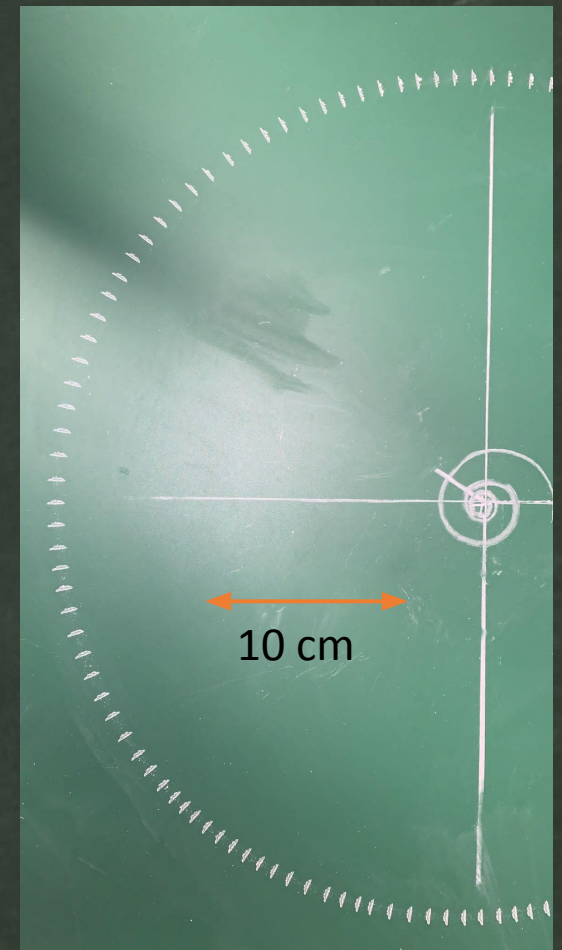
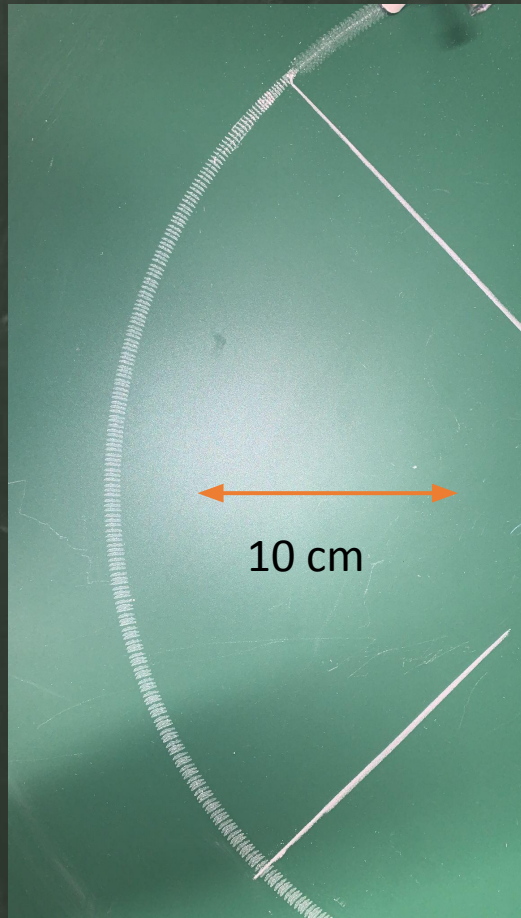
Relation tension-vitesse

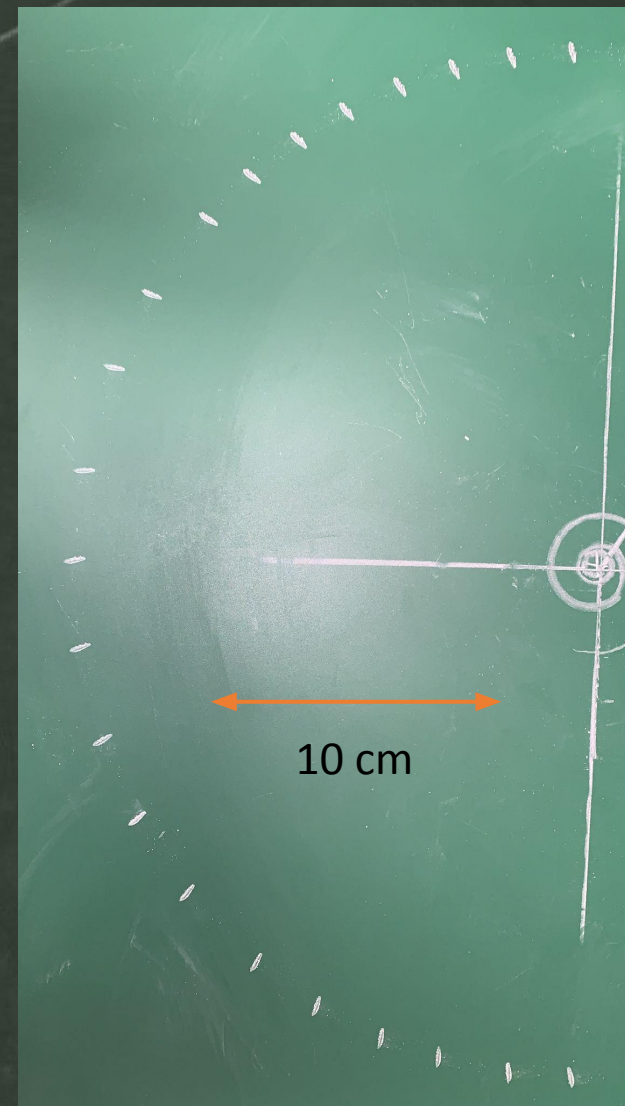
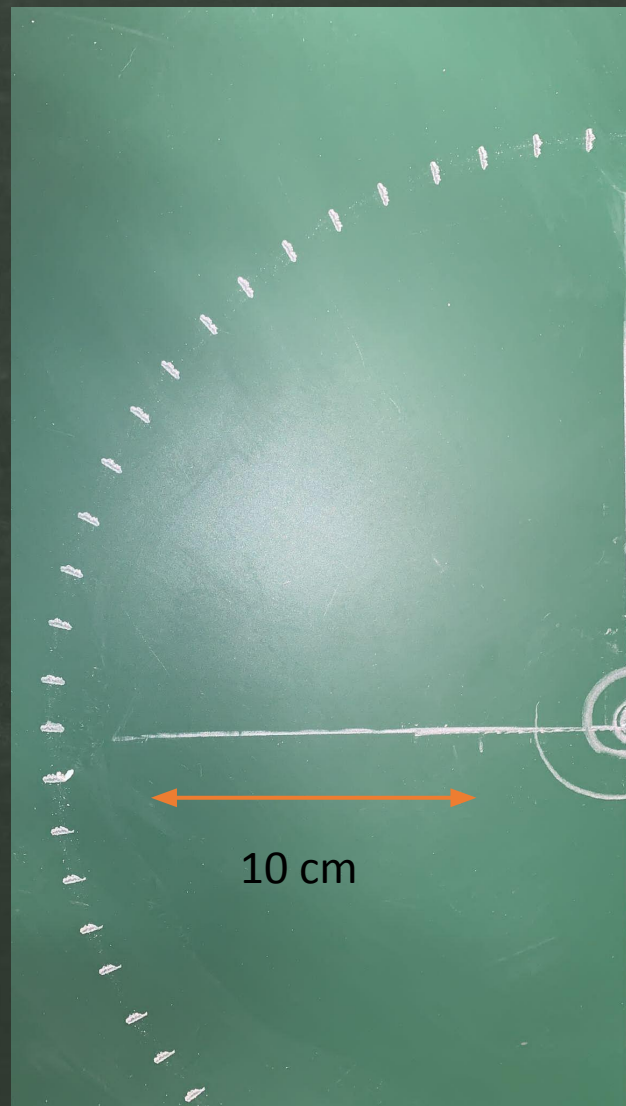
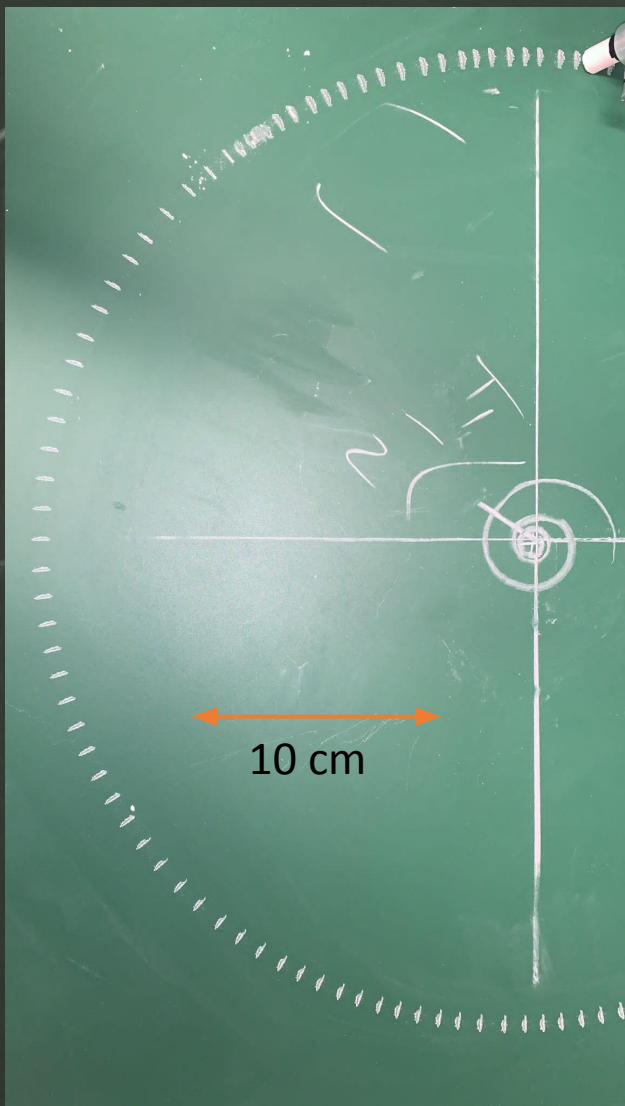
Graphique indiquant la vitesse de rotation en fonction de la tension de l'alimentation



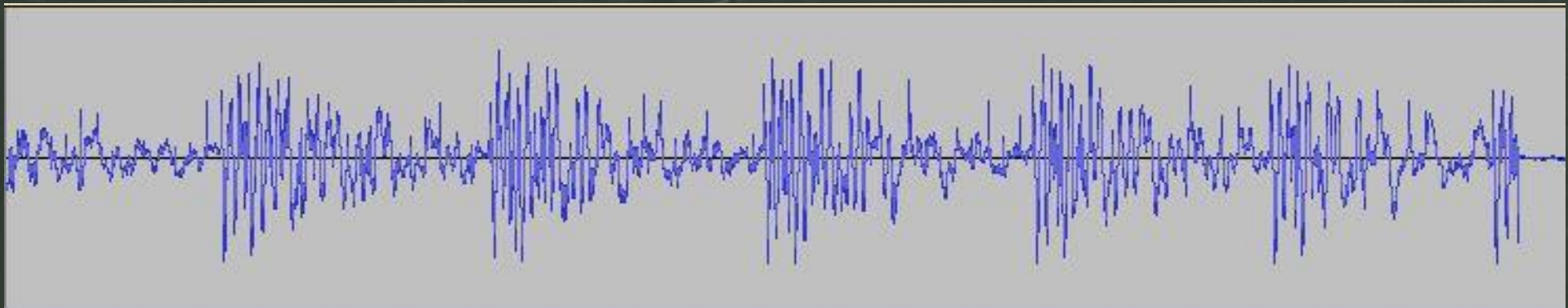
III. Resultats

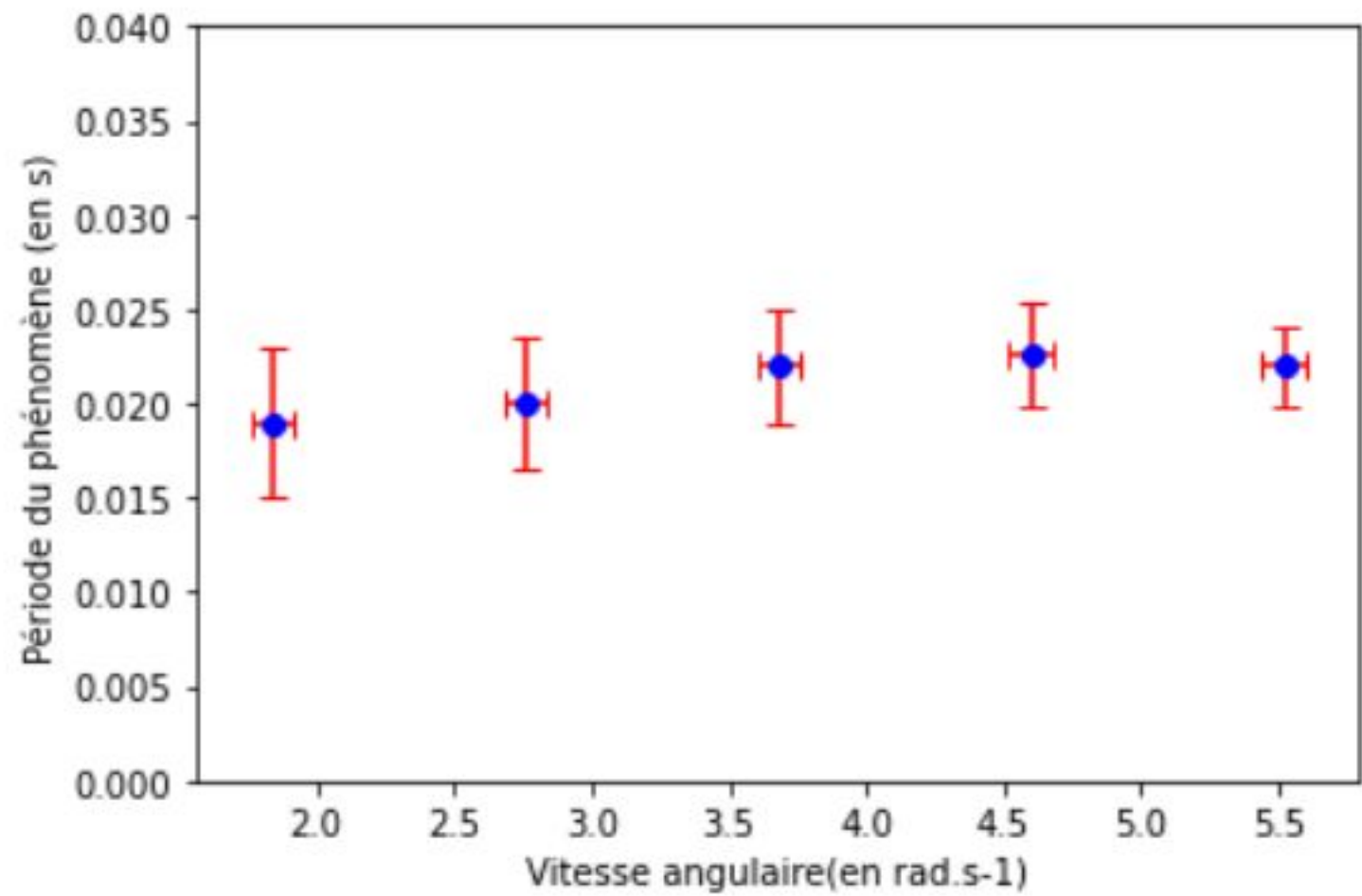
III.A – Vitesse du mouvement



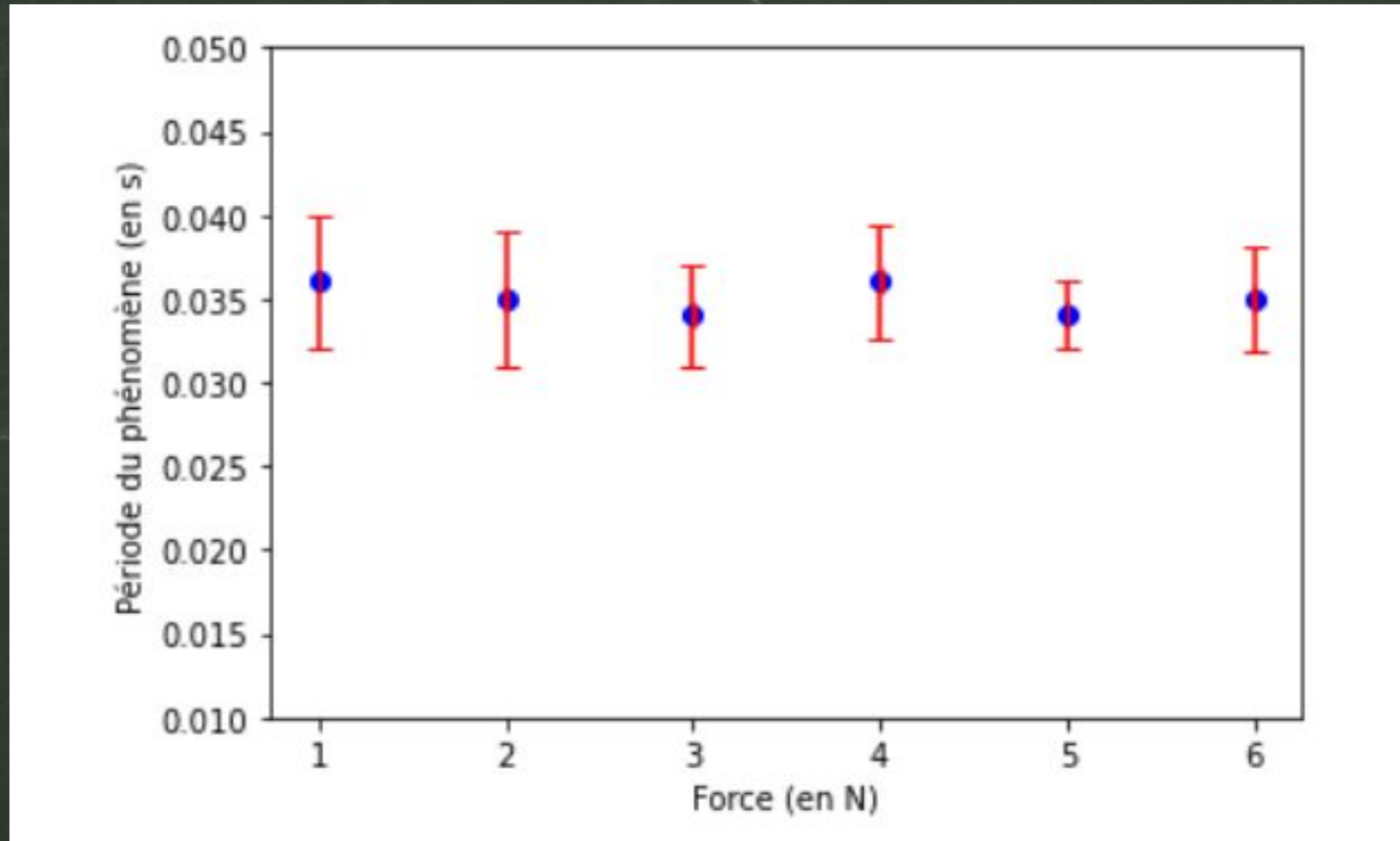


Passage à la mesure acoustique

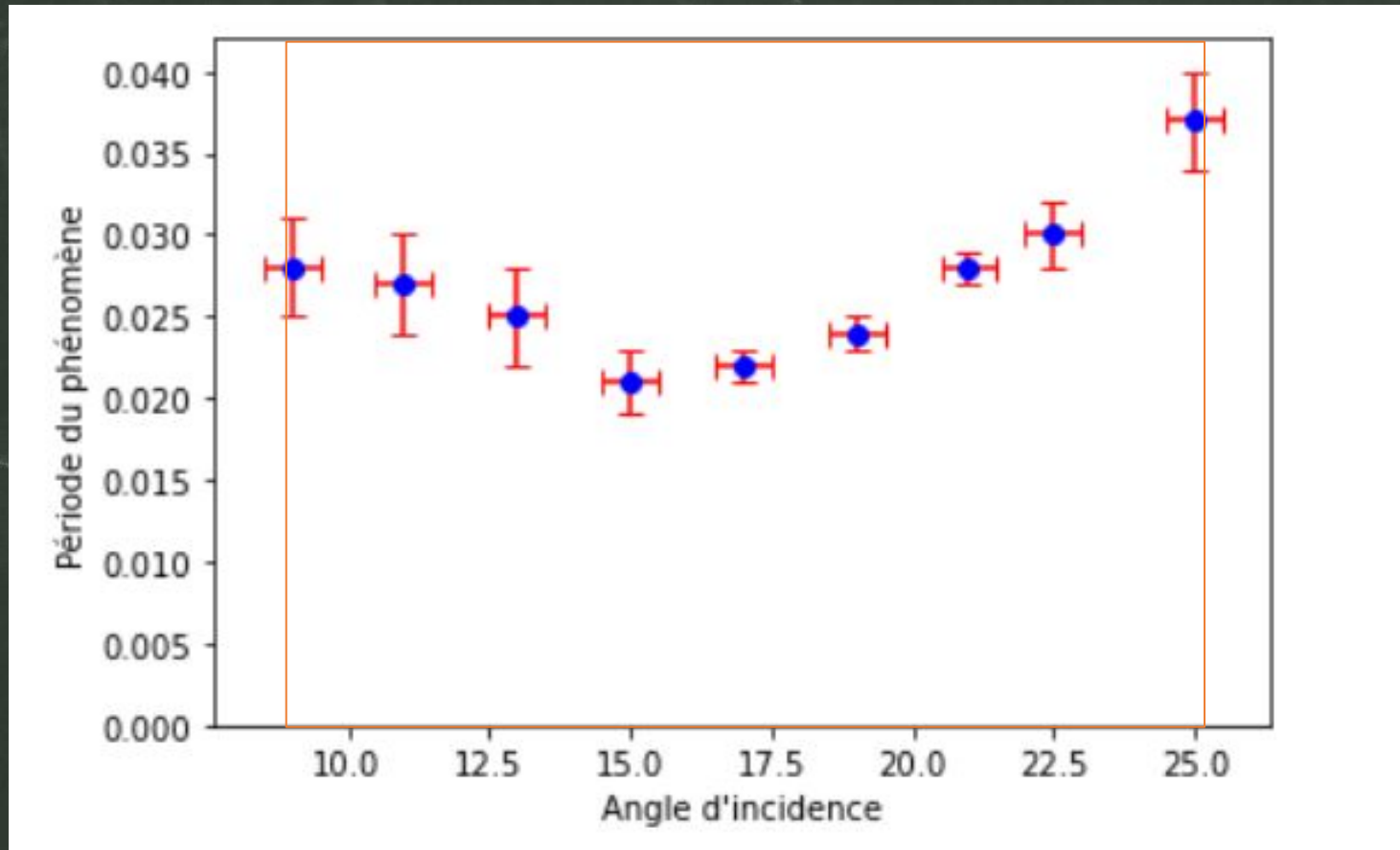




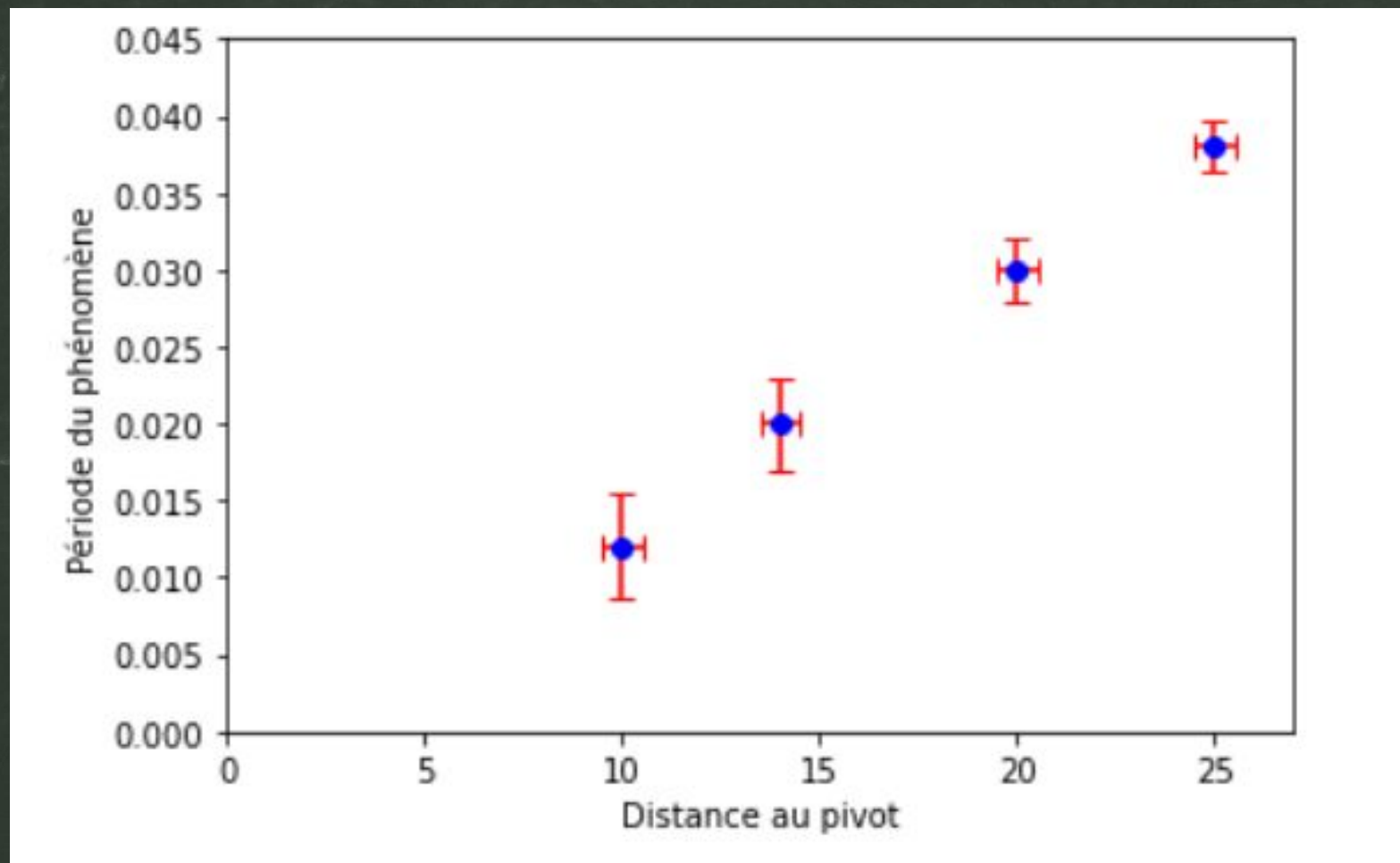
III.B – Force appliquée



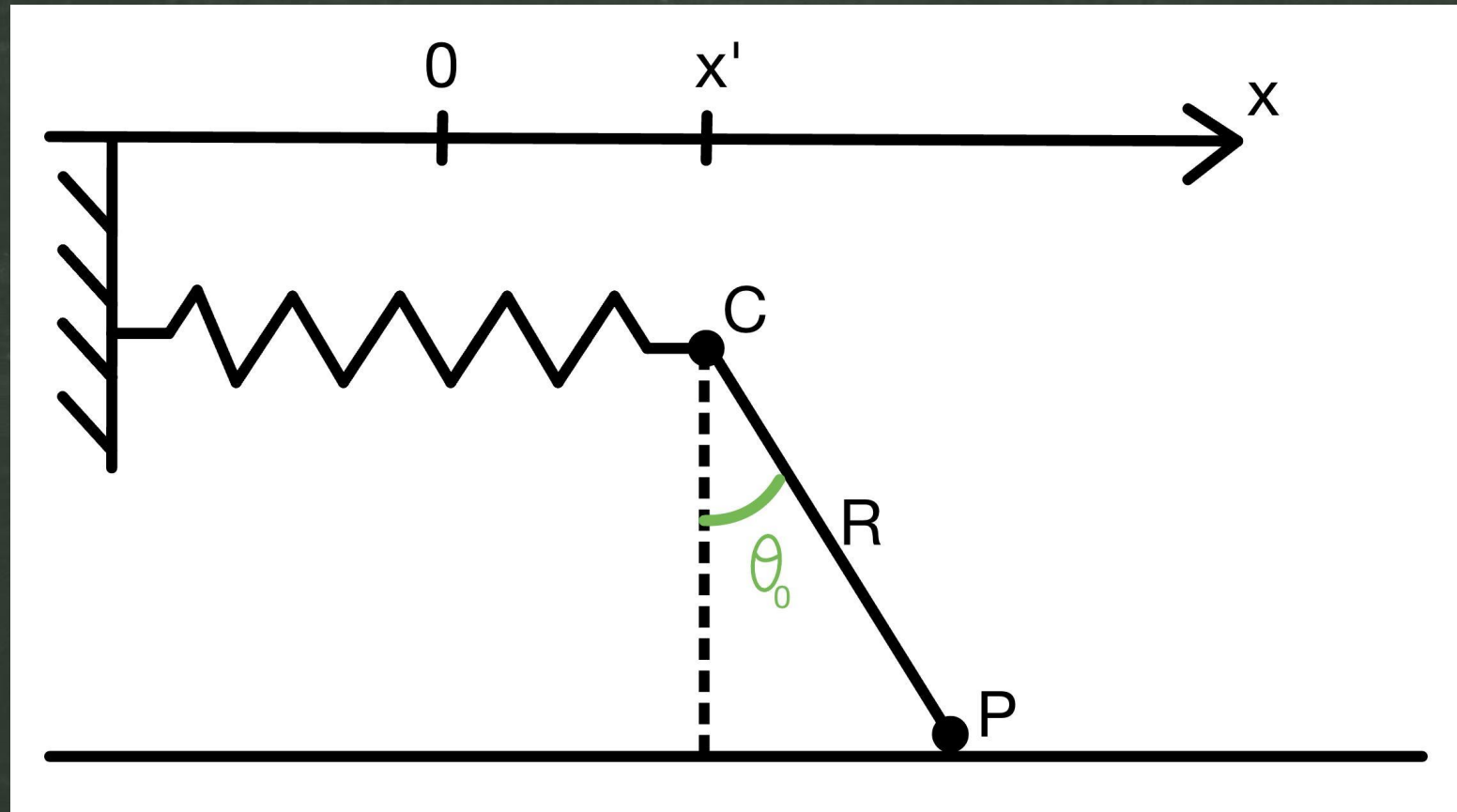
III.C – Angle de contact

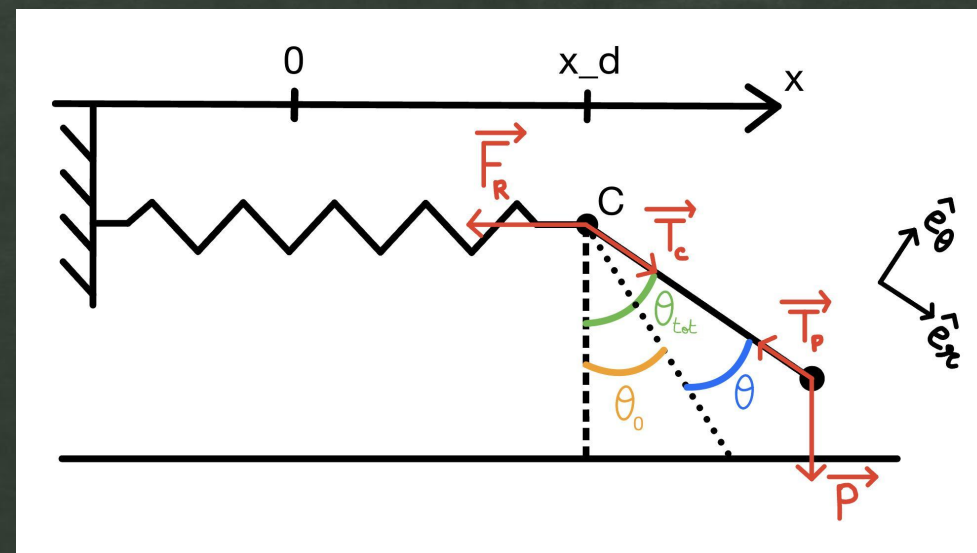
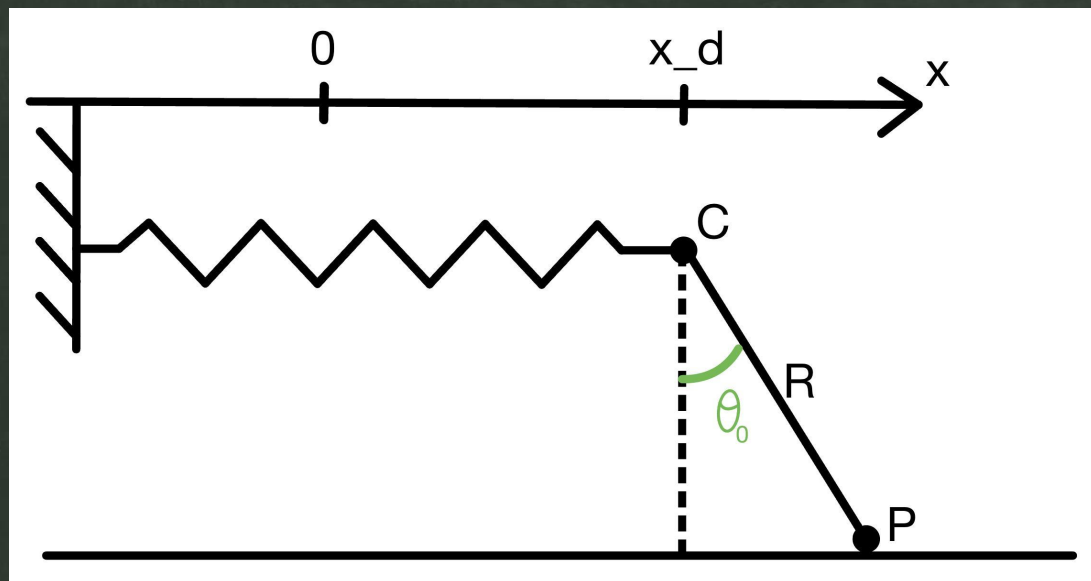
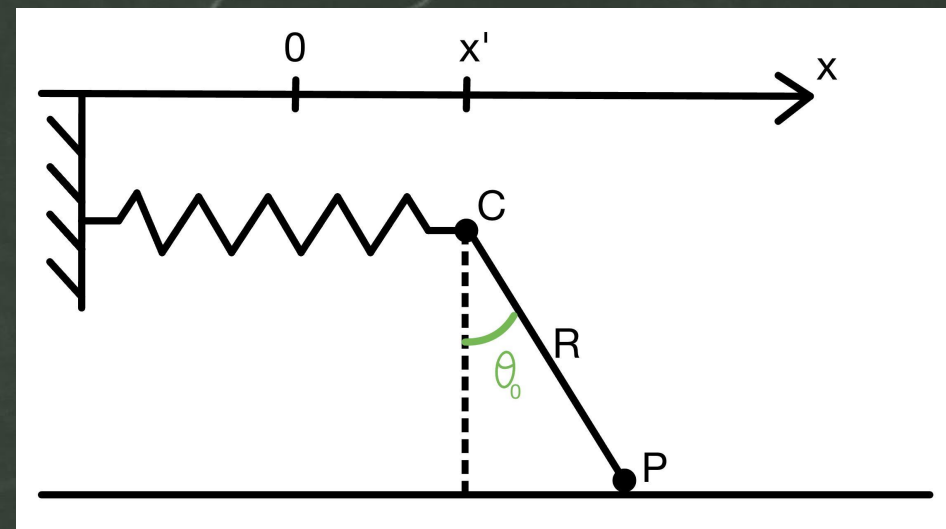
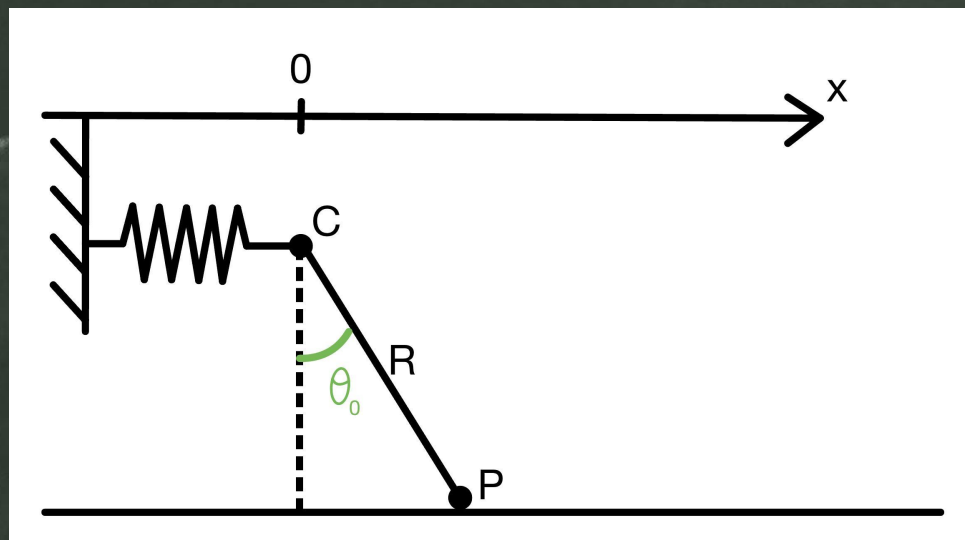


III.D – Distance au pivot



IV. Éléments de théorie





$$\tan(\theta_0) < \frac{v \omega_R}{g}$$

$$x_d = \frac{(m_1 + m_2)g \tan(\theta_0)}{k}$$

$$t_d = \frac{1}{\omega_R} \arcsin\left(\frac{g \tan(\theta_0)}{v \omega_R}\right)$$

$$\begin{aligned} \theta(t) = & (\theta_0 + \tan(\theta_0)) \cos(\Omega_p t) \\ & + \frac{\cos(\theta_0)}{R \omega_R} v \cos\left(\arcsin\left(\frac{g}{v \omega_R} \tan(\theta_0)\right)\right) \sin(\Omega_p t) \\ & - \tan(\theta_0) \end{aligned}$$

